AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

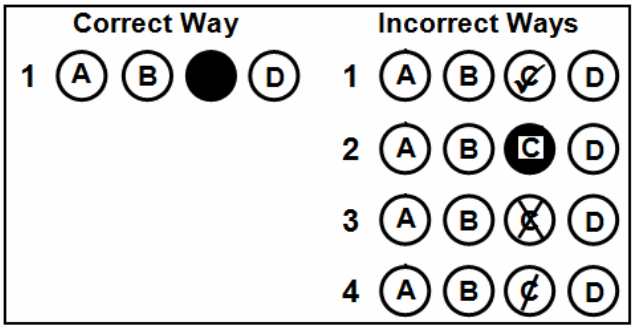
MAY 2013

Physics Paper I

Time: 35 minutes Marks: 25

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. Which of the following pairs of units is an example of S.I Units?

   A. Metre and Celsius
   B. Metre and Kelvin
   C. Gram and Celsius
   D. Gram and Kelvin

2. Which of the following instruments is used to measure the internal diameter of a pipe?

   A. Meter rule
   B. Screw gauge
   C. Vernier calipers
   D. Measuring cylinder

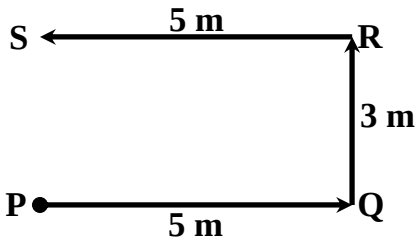
3. The number of significant figures in 204.600 is

   A. 2
   B. 3
   C. 4
   D. 6

4. In which of the following types of motion do particles of a body have the same motion?

   A. Random
   B. Rotatory
   C. Vibratory
   D. Translatory

5. The magnitude of displacement from point **P** to **S** is



   A. 3 m.
   B. 5 m.
   C. 8 m.
   D. 13 m.

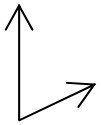
6. If a truck covers 360 m in 5 sec, its speed will be

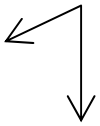
   A. 62 m/sec.
   B. 72 m/sec.
   C. 82 m/sec.
   D. 92 m/sec.

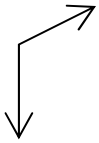
7. The given diagram represents a player kicking a football in the forward direction.



If air friction is present, then which of the following options shows the direction of forces acting on the football during the motion?

A. 

B. 

C. 

D. 

8. "Force always acts in two ways; it acts equal and opposite ways." Which of the following laws represents the above statement?

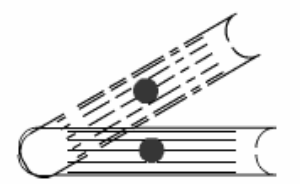
A. Newton's first law of motion
B. Newton's third law of motion
C. Law of conservation of energy
D. Law of conservation of momentum

9. A spanner is used to tighten a screw. If the torque is zero, then the applied force will be

A. at an angle of 45° with the moment arm.
B. at an angle of 270° with the moment arm.
C. in the same direction as that of the moment arm.
D. in the perpendicular direction with the moment arm.

10. The given diagram represents that the book is in the state of

A. static equilibrium.
B. stable equilibrium.
C. neutral equilibrium.
D. unstable equilibrium.



11. The stability of every body depends on the position of the centre of

    A. mass.
    B. a body.
    C. gravity.
    D. a surface.

12. If we move from the ground to the top of a mountain, then the value of ‘g’ will

    A. increase.
    B. decrease.
    C. remain same.
    D. become zero.

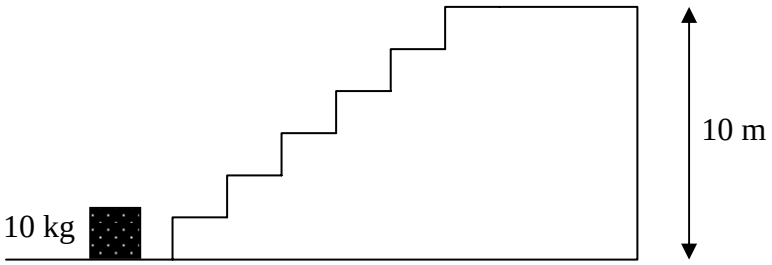
13. If there is no gravitational force of earth, then all satellites would

    A. be at rest.
    B. orbit in a circle.
    C. fall on the earth.
    D. escape from the earth’s attraction.

14. When billions of stars are held together by the force of gravity, they form a

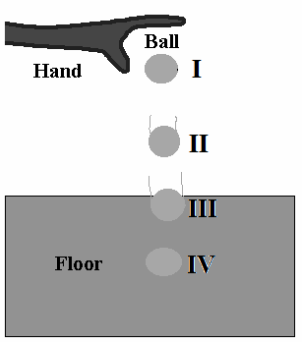
    A. comet.
    B. cluster.
    C. nebula.
    D. galaxy.

15. In the given figure, if a block is shifted from lower to higher position, then how much work will be done?



    A. 10 J
    B. 20 J
    C. 100 J
    D. 1000 J

16. A metallic ball is dropped from a height as shown in the given figure. Neglecting the air resistance, the total energy of the ball will be



A. maximum at I and II only.
B. maximum at II and III only.
C. same at II and III only.
D. same at all positions.

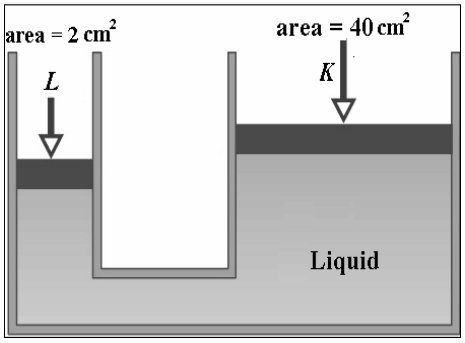
17. Which of the following options represents the energy changes taking place in a fossil fuel power station?

A. Heat → light → electrical
B. Heat → kinetic → electrical
C. Heat → chemical → electrical
D. Kinetic → electrical → potential

18. Gases are composed of large number of particles. Which of the following types of motion is performed by a gas particles?

A. Linear
B. Circular
C. Random
D. Vibratory

19. The system shown in the given diagram contains an incompressible liquid. A downward force of 80 N is exerted on the piston *K*.



What will be the upward force on piston *L*?

A. 1 N
B. 4 N
C. 80 N
D. 1600 N

20. Force acting on unit area of an object is called

A. strain.
B. stress.
C. viscosity.
D. elasticity.

21. The bulb in the clinical thermometer is made of thin glass so that

    A. the digits can be read easily.
    B. mercury can be seen clearly.
    C. the patient can feel it light in weight.
    D. mercury responds quickly to changes in temperature.

22. The process of conversion of a liquid into gaseous state at room temperature is called

    A. boiling.
    B. convection.
    C. evaporation.
    D. condensation.

23. Equal volumes of copper metal, water and air were heated equally. Which of the following options shows the correct sequence of the increasing order of expansion?

    A. Copper, water, air
    B. Water, air, copper
    C. Air, copper, water
    D. Air, water, copper

24. The process of heat transfer by collision of molecules in solids is called

    A. radiation.
    B. convection.
    C. conduction.
    D. evaporation.

25. Which of the following is TRUE about the requirement of medium in the three processes of heat transfer?

	Material where Medium is Required	**Material where Medium is NOT Required**
A	Conduction	Convection
B	Convection	Conduction
C	Conduction	Radiation
D	Radiation	Conduction

END OF PAPER

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